

Asymptotic Notation, Phase Lines, and Stability

Tuesday, September 1, 2026

Notation: Big-O notation near a finite point

Let $D \subseteq \mathbb{R}^m$, let $\mathbf{a} \in \mathbb{R}^m$, and suppose that D contains points arbitrarily close to $\mathbf{a}$ other than $\mathbf{a}$. Let $g : D \rightarrow \mathbb{R}^p$ and $h : D \rightarrow \mathbb{R}$. The statement

$$g(\mathbf{x}) = O(h(\mathbf{x})) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

means that there are constants $C > 0$ and $\delta > 0$ such that

$$\|g(\mathbf{x})\| \leq C|h(\mathbf{x})|$$

whenever $\mathbf{x} \in D$ and

$$0 < \|\mathbf{x} - \mathbf{a}\| < \delta.$$

The domain D specifies any one-sided restriction. For example, when $m = 1$, the choices $D = (0, \infty)$ and $a = 0$ specify $x \downarrow 0$.

Notation: Big-O notation at infinity

Let $D \subseteq \mathbb{R}^m$ be unbounded, let $g : D \rightarrow \mathbb{R}^p$, and let $h : D \rightarrow \mathbb{R}$. The statement

$$g(\mathbf{x}) = O(h(\mathbf{x})) \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D)$$

means that there are constants $C > 0$ and $M > 0$ such that

$$\|g(\mathbf{x})\| \leq C|h(\mathbf{x})|$$

whenever $\mathbf{x} \in D$ and $\|\mathbf{x}\| > M$. For a scalar argument, $x \rightarrow +\infty$ uses $x > M$, whereas $x \rightarrow -\infty$ uses $x < -M$.

Observation: An approach label does not assert a limit

In a statement such as

$$g(x) = O(h(x)) \quad (x \rightarrow a),$$

the label $x \rightarrow a$ specifies where the estimate must hold. It does not assert that $g(x)$, $h(x)$, or their quotient has a limit as $x \rightarrow a$.

Notation: Little-o notation near a finite point

Let $D \subseteq \mathbb{R}^m$, let $\mathbf{a} \in \mathbb{R}^m$, and let $g : D \rightarrow \mathbb{R}^p$ and $h : D \rightarrow \mathbb{R}$. Suppose D contains points arbitrarily close to $\mathbf{a}$ other than $\mathbf{a}$. The statement

$$g(\mathbf{x}) = o(h(\mathbf{x})) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

means that, for every $\eta > 0$, there is a $\delta > 0$ such that

$$\|g(\mathbf{x})\| \leq \eta |h(\mathbf{x})|$$

whenever $\mathbf{x} \in D$ and $0 < \|\mathbf{x} - \mathbf{a}\| < \delta$.

Observation: Ratio test for little-o notation

Let $D \subseteq \mathbb{R}^m$, let $\mathbf{a} \in \mathbb{R}^m$, let $g : D \rightarrow \mathbb{R}^p$, and let $h : D \rightarrow \mathbb{R}$. Suppose D contains points arbitrarily close to $\mathbf{a}$ other than $\mathbf{a}$, and suppose there is a $\delta_0 > 0$ such that $h(\mathbf{x}) \neq 0$ whenever $\mathbf{x} \in D$ and $0 < \|\mathbf{x} - \mathbf{a}\| < \delta_0$. Then

$$g(\mathbf{x}) = o(h(\mathbf{x})) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

if and only if

$$\frac{\|g(\mathbf{x})\|}{|h(\mathbf{x})|} \rightarrow 0 \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D).$$

Notation: Little-o notation at infinity

Let $D \subseteq \mathbb{R}^m$ be unbounded, let $g : D \rightarrow \mathbb{R}^p$, and let $h : D \rightarrow \mathbb{R}$. The statement

$$g(\mathbf{x}) = o(h(\mathbf{x})) \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D)$$

means that, for every $\eta > 0$, there is an $M > 0$ such that

$$\|g(\mathbf{x})\| \leq \eta |h(\mathbf{x})|$$

whenever $\mathbf{x} \in D$ and $\|\mathbf{x}\| > M$. For a scalar argument, $x \rightarrow +\infty$ uses $x > M$, whereas $x \rightarrow -\infty$ uses $x < -M$.

Notation: Asymptotic equivalence near a finite point

Let $D \subseteq \mathbb{R}^m$, let $\mathbf{a} \in \mathbb{R}^m$, and let $g, h : D \rightarrow \mathbb{R}$. Suppose D contains points arbitrarily close to $\mathbf{a}$ other than $\mathbf{a}$, and suppose there is a $\delta_0 > 0$ such that $h(\mathbf{x}) \neq 0$ whenever $\mathbf{x} \in D$ and $0 < \|\mathbf{x} - \mathbf{a}\| < \delta_0$. The statement

$$g(\mathbf{x}) \sim h(\mathbf{x}) \quad (\mathbf{x} \rightarrow \mathbf{a}, \mathbf{x} \in D)$$

means that

$$\lim_{\substack{\mathbf{x} \rightarrow \mathbf{a} \\ \mathbf{x} \in D}} \frac{g(\mathbf{x})}{h(\mathbf{x})} = 1.$$

Equivalently, $g(\mathbf{x}) - h(\mathbf{x}) = o(h(\mathbf{x}))$ along the same approach.

Notation: Asymptotic equivalence at infinity

Let $D \subseteq \mathbb{R}^m$ be unbounded, let $g, h : D \rightarrow \mathbb{R}$, and suppose $h(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in D$ with sufficiently large norm. The statement

$$g(\mathbf{x}) \sim h(\mathbf{x}) \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D)$$

means that

$$\frac{g(\mathbf{x})}{h(\mathbf{x})} \rightarrow 1 \quad (\|\mathbf{x}\| \rightarrow \infty, \mathbf{x} \in D).$$

For a scalar argument, $x \rightarrow +\infty$ and $x \rightarrow -\infty$ use the corresponding one-sided tails.

Definition: Phase line

A scalar autonomous system on an interval $I \subseteq \mathbb{R}$ has the form

$$\dot{x} = f(x), \quad x \in I.$$

Its **phase line** consists of the equilibria $f(x^*) = 0$ together with right arrows where $f(x) > 0$ and left arrows where $f(x) < 0$.

Visualization: From the vector field to the phase line

condition at x	$\dot{x}$	motion
$f(x) > 0$	$\dot{x} > 0$	$\rightarrow$
$f(x) = 0$	$\dot{x} = 0$	$\bullet$
$f(x) < 0$	$\dot{x} < 0$	$\leftarrow$

Observation: Arrows between equilibria

If f is continuous and an interval contains no equilibrium, then f has constant sign there. Every solution that remains in that interval is strictly monotone.

Definition: Lyapunov stability

Consider an autonomous system on an interval $I \subseteq \mathbb{R}$ with unique forward solutions. An equilibrium $x^* \in I$ is **Lyapunov stable** if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that every $x_0 \in I$ with $|x_0 - x^*| < \delta$ generates a solution defined for all $t \geq 0$ and

$$|x(t; x_0) - x^*| < \varepsilon \quad \text{for every } t \geq 0.$$

It is **unstable** if it is not Lyapunov stable.

Definition: Attracting equilibrium

Consider an autonomous system on an interval $I \subseteq \mathbb{R}$ with unique forward solutions. An equilibrium $x^* \in I$ is **attracting** if some $\eta > 0$ has the following property: whenever $x_0 \in I$ and $|x_0 - x^*| < \eta$, the solution is defined for all $t \geq 0$ and

$$\lim_{t \rightarrow \infty} x(t; x_0) = x^*.$$

Definition: Basin of attraction

For an autonomous system with unique solutions on an interval $I \subseteq \mathbb{R}$, the **basin of attraction** of an attracting equilibrium $x^* \in I$ is

$$\mathcal{B}(x^*) = \left\{ x_0 \in I : \begin{array}{l} x(t; x_0) \text{ exists for every } t \geq 0, \\ \lim_{t \rightarrow \infty} x(t; x_0) = x^* \end{array} \right\}.$$

Definition: Asymptotic stability

An equilibrium is **asymptotically stable** if it is both Lyapunov stable and attracting.

Question: What does the phase line of the sine flow determine?

For

$$\dot{x} = \sin x,$$

find and classify every equilibrium. For $x(0) = \pi/4$, determine the direction of motion, the limiting state, and the concavity of $x(t)$.

Solution. The equilibria are

$$x^* = k\pi, \quad k \in \mathbb{Z}.$$

For $j \in \mathbb{Z}$,

$$\sin x \begin{cases} > 0, & x \in (2j\pi, (2j+1)\pi), \\ < 0, & x \in ((2j-1)\pi, 2j\pi), \end{cases}$$

so the phase line is

$$\cdots \longrightarrow (-\pi) \longleftarrow (0) \longrightarrow (\pi) \longleftarrow (2\pi) \longrightarrow \cdots.$$

Thus $x^* = (2j+1)\pi$ is asymptotically stable and $x^* = 2j\pi$ is unstable.

For $x(0) = \pi/4$, the solution increases and cannot cross either equilibrium 0 or π . Hence it remains in $(0, \pi)$ and is bounded above, so it has a limit $L \in [\pi/4, \pi]$ as $t \rightarrow \infty$. If $L < \pi$, then $\sin L > 0$.

Question: What does the phase line of the sine flow determine? (continued)

Continuity would give constants $c > 0$ and $t_0 > 0$ such that $\dot{x}(t) = \sin x(t) \geq c$ for every $t \geq t_0$, contradicting convergence to the finite number L . Therefore

$$\lim_{t \rightarrow \infty} x(t) = \pi.$$

Differentiating $\dot{x} = \sin x$ with respect to t gives

$$\ddot{x} = \cos x \dot{x} = \sin x \cos x.$$

Therefore

$$\ddot{x} \begin{cases} > 0, & 0 < x < \pi/2, \\ = 0, & x = \pi/2, \\ < 0, & \pi/2 < x < \pi. \end{cases}$$

The time graph is increasing, initially concave up, and then concave down.

Question: Can an implicit equilibrium be counted and classified?

Analyze

$$\dot{x} = x - \cos x$$

without solving the equilibrium equation in closed form.

Solution. An equilibrium satisfies $x = \cos x$, so it must lie in $[-1, 1]$. Let

$$g(x) = x - \cos x.$$

On $[-1, 1]$,

$$g'(x) = 1 + \sin x > 0.$$

Also,

$$g(0) = -1 < 0, \quad g(1) = 1 - \cos 1 > 0.$$

The intermediate value theorem gives at least one zero in $(0, 1)$, and strict monotonicity gives at most one. Hence there is exactly one equilibrium $x^* \in (0, 1)$. Numerically,

$$g(0.739) < 0, \quad g(0.740) > 0, \quad x^* \approx 0.739085.$$

Because $g(x) < 0$ immediately to the left of x^* and $g(x) > 0$ immediately to the right, both adjacent arrows point away from x^* . Thus x^* is unstable.